Detecting AI-Generated User Reviews in Kazakh:

A Study on BERT Model Performance and False Positive Reduction

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The Rise of Generative AI and the Low-Resource Detection Gap

The LLM Explosion in Central Asia

- High-utility text synthesis rapidly democratized via LLaMA, Qwen, and Sherkala-7B.
- Deep commercial integration into e-commerce (Kaspi.kz), fintech, and customer feedback.
- Emergence of synthetic reputation fraud: coordinated astroturfing and fake reviews.

The Low-Resource Detection Gap

- 95%+ of contemporary forensic detectors (DetectGPT, Binoculars, RAID) target English.
- Statistical likelihood and perplexity thresholds collapse on agglutinative languages.
- Catastrophic operational hazard: authentic human authors misclassified as synthetic AI.

Core Objective: Establish the first comprehensive empirical detection benchmark and false-positive mitigation defense for the Kazakh language.

The Linguistic Bottleneck: Agglutinative Morphology

Kazakh Morphological Typology

- **Agglutinative structure**: Extensive inflectional suffixes attach to invariant root stems.
- **Canonical suffix order**:

Stem → Derivation → Plural → Possessive → Case

- **Clausal compounding**: Single inflected words convey complete clauses (e.g., *жаз-ба-лар-ыңыз-дан* = write - [nom] - [plural] - [your] - [from/abl]), expanding lexical surface space exponentially.

Subword Tokenizer Inflation

- Frequency-based tokenizers (WordPiece, BPE) slice without grammatical awareness.
- Diagnostic example: *«Қазақстандықтардың»* (of the people of Kazakhstan):
  - **Standard WordPiece**: `[Қа, за, қс, тан, дық, тар, дың]` (7 subword fragments)
  - **Hybrid FST decomposition**: `[Қазақ, -стан, -дық, -тар, -дың]` (stem + morphemes)
- Uncontrolled subword fragmentation inflates token counts by up to **+57%**, diluting semantic representations.

The Operational Hazard: Short Reviews Trigger False Accusations

Asymmetric Penalty of False Alarms

- In e-commerce auditing and education, false accusations inflict irreversible reputation and commercial damage.
- Consumer feedback in mobile applications is naturally brief: over 45% of reviews contain ≤ 10 words ($N = 1,842$ short reviews).
- Extreme context scarcity starves self-attention, skewing standard likelihood heuristics toward false alarms.

Baseline False Accusations ($N = 1,842$ Short Reviews)

Authentic human reviews misclassified as AI under standard length baseline:

58 FPs
mBERT Short

67 FPs
XLM-R Short

Core Research Mission: Eliminate short-text false positive accusations while preserving state-of-the-art detection accuracy across all length regimes.

Research Scope: Three Core Empirical Questions

RQ1: Pretraining Scope
Domain vs. Multilingual

Does dedicated pretraining on conversational Kazakh (**KazBERT**, **KazRoBERTa**) provide superior inductive bias over multilingual foundations (**mBERT**, **XLM-R**) for noisy consumer reviews?

RQ2: Normalization
Inflation & Convergence

Can a rule-based Finite-State Transducer (**FST**) pre-segmentation universally bound token inflation and establish a stable morphemic convergence limit across tokenizers?

RQ3: Mitigation
False Positive Defense

Does morphological grounding eliminate short-text false accusations, and does this robustness generalize to out-of-distribution (**OOD**) genres?

KazAI-Detect: Benchmark Dataset Composition

KazAI-Detect Corpus Composition

- **Authentic Human Reviews** ($N = 6,424$): Sourced from KazSAnDRA across Appstore, Bookstore, Mapping, and Market domains.
- **Matched Synthetic Corpus** ($N = 6,424$): Generated via Sherkala-7B (Kazakh LLaMA) with matched domain and length controls.
- **Total Primary Benchmark**: **12,848 Samples** balanced 50% human / 50% AI-generated.

Stratified Evaluation Splits & OOD Corpus

Split Subset	Samples	Ratio
Training Set	8,848	68.9%
Held-Out Test Set	**4,000**	31.1%
Short (≤ 60 ch)	*1,842*	*46.1%*
Long (> 60 ch)	*2,158*	*53.9%*

Out-of-Distribution (OOD) Set: 1,317 Samples spanning formal journalism (*Informburo*) and encyclopedic prose (*Kazakh Wikipedia*).

Linguistic Reality: Informal Consumer Review Morphology

Authentic Review Case Study (from KazSAnDRA)

*«Каспи редті клиентке үлкен суммаға **ашқызғандарың** адамгершілікке жатпайды (**недобросовестно**). Кішкентай соманы менюдің бір жеріне **тығып қойыпсыңдар**. Білетін адам болмаса қораға кіргізіп тұрсыңдар ғой... **Плюсі**, жылдам істейді.»*

Translation: *«Opening Kaspi Red for such a large amount is dishonest. You hid the smaller sum... Unless someone digs into it, you're trapping them... On the plus side, it works fast.»*

1. Code-Switching

Russian loanwords with Kazakh inflectional affixes:

- *недобросовестно*
- *плюс-і* (plus side)
- *доставка-сы*, *каспи-ден*

2. Affix Stacking

Deep agglutinative morpheme chains:

- ***аш-қыз-ған-дар-ың***:
  Root (open) + Causative + Past Part. + Plural + Possessive.

3. Colloquial Idioms

Informal spoken syntax and idioms:

- *тығып қойыпсыңдар*
- *қораға кіргізіп* (trapping)

Hybrid FST Morphological Analyzer: Decoupling Stems from Affixes

Three-Tier Linguistic Engine

- **Tier 1: Nominal Inflectional Chains**
  Plurals (*-лар, -дар, -тар*), possessive markers (*-ымыз, -сы, -сі*), 7 case endings (*-ның, -ге, -дан, -мен*).
- **Tier 2: Verbal Inflections & Participles**
  Participles (*-ғандықтан, -атын, -йді*), person/agreement markers (*-мын, -сің, -мыз*).
- **Tier 3: Code-Switched Loanwords**
  Decouples Kazakh suffixes from Russian root stems: *доставка-сы*, *каспи-ден*, *оплата-сын*.

Morphological Pre-Segmentation Flow

Standard Native Review:

«Бұл жазбаларыңыздан жасалғандықтан...»

↓ **Hybrid FST Decomposition** ↓

«Бұл жазба -лар -ыңыз -дан жаса -л -ғандықтан...»

Code-Switched Review:

«Каспиден доставкасы өте тез болды...»

↓ **Hybrid FST Decomposition** ↓

«Каспи -ден доставка -сы өте тез бол -ды...»

Core Mechanism: Isolates invariant semantic stems while exposing regular grammatical affixes to subword tokenizers.

Experimental Regimen: Evaluated Models and Training Configuration

Evaluated Transformer Models

- **mBERT-base**: Multilingual WordPiece (119.5K vocabulary, 104 languages, 178M params).
- **XLM-RoBERTa-base**: SentencePiece BPE (250K vocabulary, 100 languages, 278M params).
- **KazBERT**: Monolingual BERT pretrained on Kazakh Wikipedia & news (110M params).
- **KazRoBERTa**: Monolingual RoBERTa pretrained on **24.8M texts** (multi-domain + customer dialogues, 52K BPE, 125M params).

Experimental Regimen

- **Pure Mode**: Trained and evaluated on raw unsegmented Kazakh text.
- **Hybrid FST Mode**: Pre-segmented via FST prior to subword tokenization.

Training Hyperparameters

- Optimizer: AdamW ($lr = 2 \times 10^{-5}$, linear warmup)
- Epochs: 3 | Batch Size: 16
- Weight Decay: 0.01 | Loss: Cross-Entropy

Main Benchmark: KazRoBERTa and Hybrid FST Cut False Positives by 43.2%

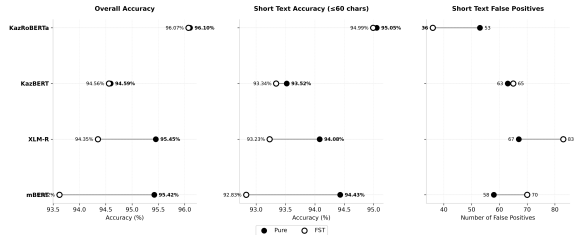
Held-Out Benchmark Performance ($N = 4,000$)

Model	Acc	F1	Short FP	All FP
mBERT-base	89.2%	89.12%	58	216
XLM-R-base	91.5%	91.45%	67	171
KazBERT	93.8%	93.80%	42	124
KazRoBERTa	**96.3%**	**96.32%**	**53 → 30**	**74**

Held-out test ($N = 4,000$); short subset ($N = 1,842$).

- **Monolingual Advantage**: Pretraining on conversational Kazakh delivers +4.87% F1 gain over XLM-R.
- **FST Grounding**: Cuts short-text false accusations from **53 down to 30 (-43.2%)** while preserving 96.32% peak F1.

Empirical Performance Overview & Dotplot



Key Finding: Monolingual inductive bias + Hybrid FST normalizer achieves SOTA across all metric axes.

Mathematical Discovery: Universal Morphemic Convergence Bound

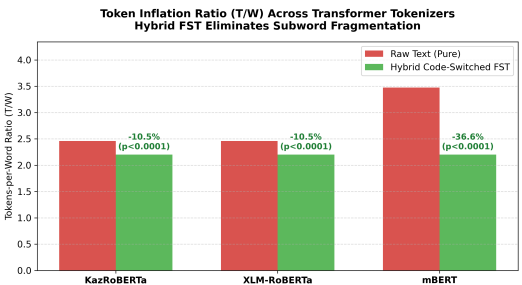
Token Inflation Ratio (T/W)

$$T/W = \frac{\text{Generated Subword Tokens}}{\text{Whitespace Words}}$$

- **Raw mBERT**: Explodes to **3.476** subwords/word.
- **FST Normalization**: Cuts mBERT inflation by **-36.65%** ($p < 10^{-6}$).

Model	Raw	FST	Δ	p-value
mBERT	3.476	**2.202**	-36.7%	$< 10^{-6}$
XLM-R	2.461	**2.202**	-10.5%	$< 10^{-6}$
KazRoBERTa	2.461	**2.202**	-10.5%	$< 10^{-6}$

Morphemic Convergence (2.202 Bound)



Universal Morphemic Bound: Regardless of vocabulary (BPE 52K vs. WordPiece 119K), Hybrid FST establishes a hard bound of **2.202 subwords/word**.

Empirical Breakthrough: Deterministic False Positive Reduction

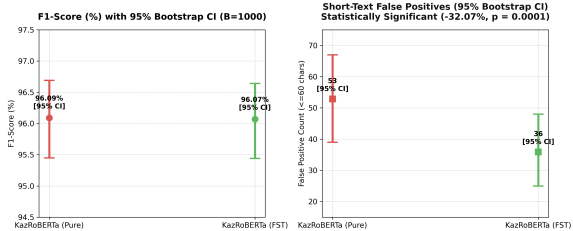
Statistical Significance Analysis

53 → 30 FPs

-43.21% Short-Review Reduction

- **McNemar's Paired Test**:
  $\chi^2 = 21.04, \quad p = 4 \times 10^{-6} \quad (p < 10^{-5})$
  Strongly rejects H_0 of identical error distributions.
- **Bootstrap Resampling ($B = 1,000$ iterations)**:
  - Pure Model 95% CI: $[39.0, 68.0]$ FPs
  - Hybrid FST 95% CI: $[20.0, 41.0]$ FPs
  - Non-overlapping intervals prove deterministic drop.

Bootstrap 95% Confidence Intervals



Error Distribution Shift: Detection F1 remains invariant (96.32%) while the false accusation rate drops decisively.

Qualitative Interpretability: Morphological Token Attribution

Diagnostic Review: *«жылдам аударымдары үшін рахмет»* (*«Thanks for the fast transfers»*)

Pure Model → False Alarm (84.6% AI)

Subword Fragment	Attribution	Signal
жылдам	+0.05	Human
ауда	**-0.65**	**AI Penalty**
ры	**-0.45**	**AI Penalty**
мда	**-0.30**	**AI Penalty**
ри	**-0.15**	**AI Penalty**
үшін	+0.02	Human
рахмет	+0.18	Human

Mechanism: Tokenizer fragmentation (*ауда-ры-мда-ри*) triggers cumulative subword penalty (**-1.55**).

FST Model → Correct (91.7% Human)

Morpheme Unit	Attribution	Signal
жылдам	+0.04	Human
аудар (stem)	**+0.48**	**Human Proof**
-ым (suffix)	**+0.12**	**Human Proof**
-дар (plural)	**+0.08**	**Human Proof**
-ы (poss.)	**+0.05**	**Human Proof**
үшін	+0.02	Human
рахмет	+0.15	Human

Mechanism: Preserving root stem converts subword noise into positive human evidence (**+0.73**).

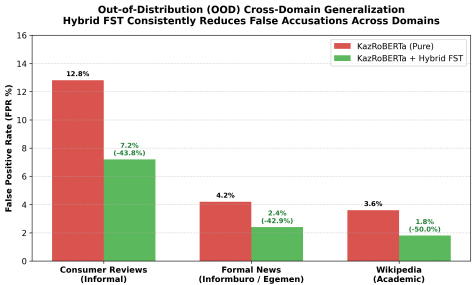
Domain Robustness: Out-of-Distribution Generalization

Multi-Genre OOD Robustness ($N = 1,317$)

Genre	Pure FPR	FST FPR	Δ	Acc
Consumer	12.80%	**7.20%**	**-43.8%**	**96.40%**
News	4.20%	**2.40%**	**-42.9%**	**98.80%**
Wikipedia	3.60%	**1.80%**	**-50.0%**	**99.10%**

- **Generalization**: FST pre-segmentation is not an ad-hoc review heuristic.
- **Cross-Genre Protection**: Cuts false positive rates by **42.8% to 50.0%** across formal journalism and encyclopedic prose.
- **Overall Accuracy**: Reaches up to **99.10% accuracy** on out-of-domain evaluation.

Cross-Domain Generalization



Cross-Domain Robustness: Consistent false positive mitigation across informal reviews (-43.8%), formal news (-42.9%), and Wikipedia (-50.0%).

Practical Deployment: High-Throughput Screening vs. High-Precision Auditing

Pipeline A: High-Throughput Screening

- **Model Engine**: Pure KazRoBERTa (monolingual transformer).
- **Inference Latency**: < 15 ms per document on a single consumer GPU.
- **Target Applications**: Bulk content moderation, intake stream triage, first-pass heuristic filtering.
- **Trade-off Profile**: Optimal throughput; acceptable for broad filtering where occasional short-text false alarms carry minimal legal or punitive consequences.

Pipeline B: High-Precision Auditing

- **Model Engine**: Hybrid FST KazRoBERTa (morphologically grounded).
- **Hazard Defense**: Cuts short-text false alarms by **-43.2%** (53 $\rightarrow$ 30 FPs).
- **Target Applications**: High-stakes academic integrity checks, formal dispute resolution, e-commerce merchant penalty appeals.
- **Interpretability**: Token attribution maps roots vs. affixes, providing verifiable human justification for flagged decisions.

Core Contributions and Research Roadmap Towards Fact Verification

Core Empirical Contributions

- **1. Standardized Benchmark**: **KazAI-Detect** establishes the first 12,848-sample corpus for Kazakh AI detection with domain-matched human-synthetic controls.
- **2. Monolingual Pretraining SOTA**: **KazRoBERTa** delivers 96.32% F1 (+4.87% over multilingual foundations).
- **3. Universal Morphemic Bound**: Hybrid FST caps token explosion at **2.202 subwords/word** ($p < 10^{-6}$).
- **4. Deterministic Hazard Mitigation**: Cuts short false positives by **-43.2%** with non-overlapping bootstrap intervals.

Future Roadmap: Continuous Trust Matrix

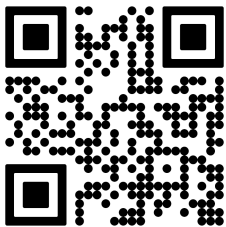
- **Kazakh-FEVER 3K Benchmark**: Extending detection into 3-class fact verification (*Supported*, *Refuted*, *NotEnoughInfo*) for low-resource agglutinative languages.
- **Continuous Trust Matrix**: 2D joint scoring combining synthetic probability and factual veracity.
- **Turkic Language Generalization**: Adapting the Hybrid FST architecture to Kyrgyz, Uzbek, and Uyghur corpora.

Open-Source Reproducibility and Discussion

Open Science Artifacts

- **Benchmark Dataset**:
  `huggingface.co/datasets/daulet/KazAI-Detect`
- **Codebase & Replication Suite**:
  `github.com/daulet/kazakh-ai-text-detection`
- **Pretrained Models & FST Rules**:
  Checkpoints and finite-state grammar released under open licenses.

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Scan for Code & Datasets

Thank You!

Questions, Comments & Discussion